Short questions

|!|BCESC0180000001|!|

Complete the following paragraph using appropriate words. (5 marks)

The musculo-skeletal system consists of (a)_____ muscles and bones. These muscles are attached to bones by (b)_____ and they work in (c)_____ pairs. When one muscle in the pair (d)_____, the other (e)_____ to move bones in a particular direction.

##

(a) skeletal (1)
(b) tendons (1)
(c) antagonistic (1)
(d) contracts* (1)
(e) relaxes* (1)
(*Accept answers in a different order)

(5 marks)

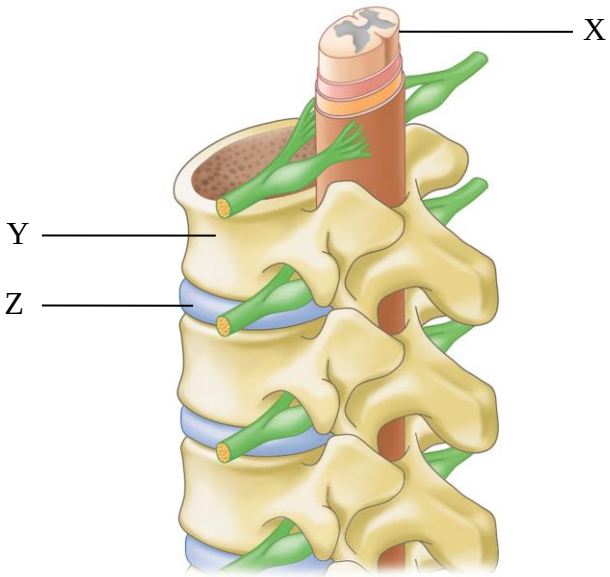
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0180000002|!| ★

The diagram below shows part of the human vertebral column.



(a) Name X, Y and Z. (3 marks)

(b) Give ***three*** functions of the vertebral column. (3 marks)

##

(a) X: spinal cord (1)
Y: vertebrae (1)
Z: intervertebral disc (1)

(b) Any ***three*** of the following: (3)

- Supporting the body weight /
- Protection of the spinal cord /
- Providing surface for muscle attachment
- Production of blood cells /
- Storage of minerals / fat

(6 marks)

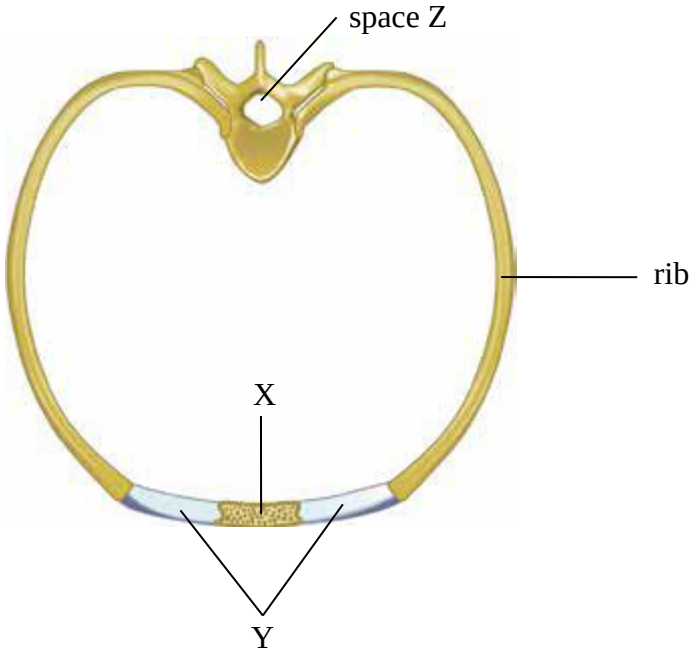
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	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	3

|!|BCESB0180000003|!| ★ **Cross topic**

The diagram below shows the top view of the rib cage.



(a) Name X and Y. (2 marks)

(b) State the structure that is enclosed in space Z. (1 mark)

(c) (i) Name the muscles associated with the ribs. (1 mark)

(ii) Describe how the muscles in (i) work together with the ribs to bring about inhalation. (3 marks)

##

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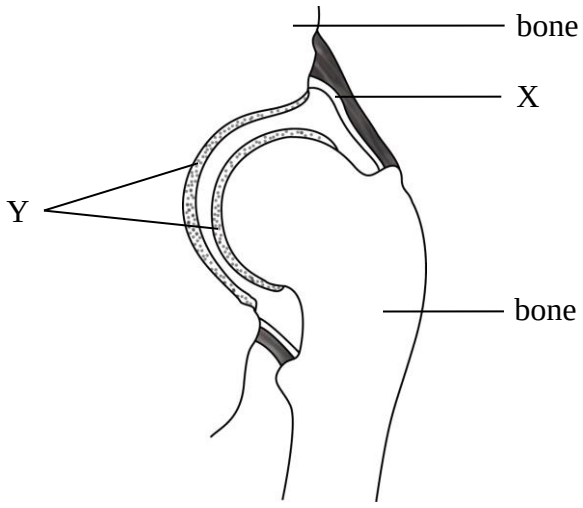
(a) X: sternum (1)
Y: cartilage (1)
(b) Spinal cord (1)
(c) (i) Intercostal muscles (1)
(ii) The intercostal muscles contract to raise the ribs upwards and outwards. (1)
This increases the volume of the thoracic cavity. (1)
Thus the air pressure in the lungs becomes lower than the atmospheric pressure. (1)
As a result, air is drawn into the lungs.

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	3

|!|BCESC0180000004|!|
The diagram below shows a movable joint.



(a) Describe the degree of movement this joint allows. (1 mark)

(b) Give ***one*** example of this type of joint in the human body. (1 mark)

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(c) Name structure X and describe its function. (2 marks)

(d) Name structure Y and describe its function. (2 marks)

##

(a) This joint allows movement of the bone in many planes / all directions. (1)

(b) Shoulder joint / Hip joint (1)

(c) Ligament (1)

It holds the bones in position to prevent dislocation during movement. (1)

(d) Cartilage (1)

It reduces friction between the bones during movement. / It acts as a shock absorber. (1)

(6 marks)

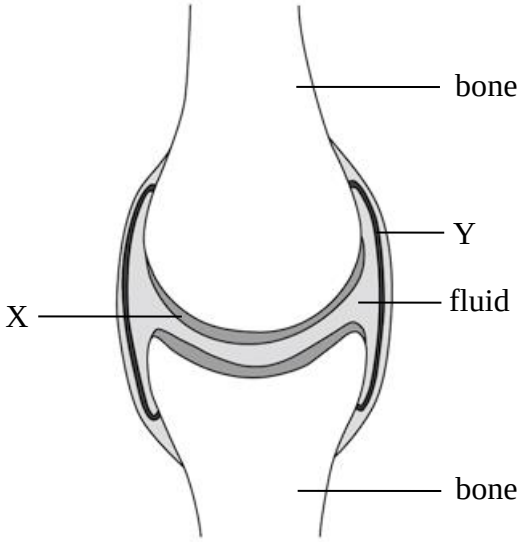
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	1
(c)	Analysing data	1
	Understanding basic concepts	1
(d)	Analysing data	1
	Understanding basic concepts	1

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|!|BCESB0180000005|!| ★

The diagram below shows a movable joint between two bones in a finger.



(a) What type of movable joint is this? Describe the degree of movement this type of joint allows. (2 marks)

(b) State ***two*** differences between tissue X and bone. (2 marks)

(c) Structure Y secrets a fluid that fills the space between in bones in the joint.

(i) Name structure Y. (1 mark)

(ii) Give ***one*** function of the fluid secreted by structure Y. (1 mark)

##

(a) Hinge joint (1)

It allows movement of bones in one plane. (1)

(b) Any ***two*** of the following: (2)

- Tissue X is softer than bone. /
- Tissue X contains fewer minerals than bone. /
- Tissue X is more elastic than bone. /
- Tissue X has no blood supply while bone has blood supply. /

(Or other reasonable answers)

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(c) (i) Synovial membrane (1)

(ii) It reduces friction between the bones during movement. / It supplies oxygen and nutrients to the cartilage. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)(i)	Analysing data	1
(c)(ii)	Understanding basic concepts	1

|!|BCESC0180000006|!|

The X-ray photograph below shows a movable joint.



(a) Identify the type of joint shown based on the degree of movement it allows. (1 mark)

(b) Describe how the bones of the joint are held together. (1 mark)

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(c) State ***two*** structures in the joint that help reduce friction during movement. Explain how they are able to carry out their functions. (4 marks)

##

(a) Hinge joint (1)

(b) The bones are held together by ligaments. (1)

(c) Cartilage (1)

It is soft and elastic; it covers the end surfaces of the bones to reduce friction. (1)

Synovial membrane (1)

It secretes synovial fluid, which lubricates the joint. (1)

(6 marks)

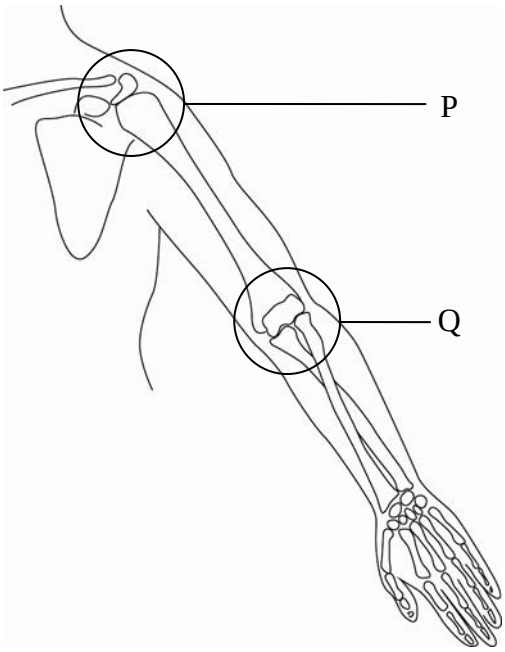
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	4

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|!|BCESC0180000007|!|

The diagram below shows the bones in the human arm.



(a) Name the types of joints found at P and Q. (2 marks)

(b) How does movement at joint P differ from that at joint Q? (2 marks)

##

(a) P: ball-and-socket joint (1)
Q: hinge joint (1)

(b) Joint P allows movement of the bones in many planes / all directions. (1)
Joint Q allows movement of the bones in one plane only. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2

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|!|BCESB0180000008|!| ★

The photographs below show two postures of a man doing sit-up exercises.



Posture 1



Posture 2

(a) Which arm muscles, biceps or triceps, are contracting when the man strengthens his arms? (1 mark)

_____

(b) Give ***two*** features of the vertebral column that allow it to bend to a curved shape as in posture 2. (2 marks)

_____

##

(a) Triceps (1)

(b) The vertebral column is made up of many vertebrae which are articulated by joints. (1)

There are compressible cartilage / intervertebral discs between the vertebrae. (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	2

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|!|BCESB0180000009|!| ★

Figure 1 shows the drawing of a human arm. Figure 2 shows the transverse section of the arm along the line AB.

Figure 1

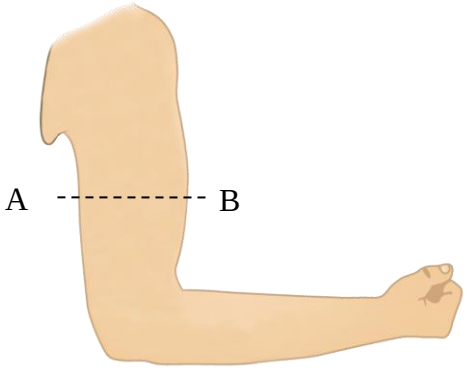
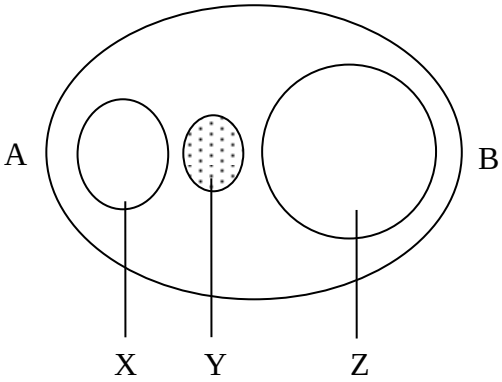


Figure 2



(a) Name muscles X and Z. (3 marks)

_____

_____

(b) Which of muscles X and Z would increase in diameter when the forearm is lowered to straighten the arm? (1 mark)

_____

(c) State ***two*** functions of Y in the arm. (2 marks)

_____

_____

_____

_____

##

(a) X: triceps (1)

Z: biceps (1)

(b) X (1)

(c) Any ***two*** of the following: (2)

- It provides surface for muscles to attach. /
- It allows movement. /
- It produces blood cells. /
- It stores minerals.

_____

(5 marks)

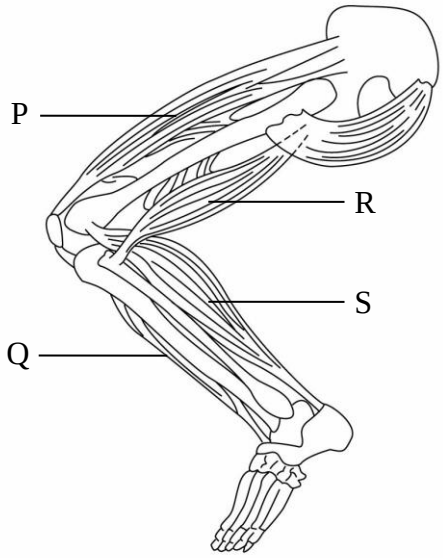
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##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	1
(c)	Understanding basic concepts	2

|!|BCESC0180000010|!|

The diagram below shows some of the bones and muscles in a human leg.



(a) Using the letters in the diagram, indicate

(i) the muscle which contracts to bend the leg at the knee. (1 mark)

(ii) the muscle contracts to straighten the leg at the knee. (1 mark)

(b) Name the structures that connect muscles to bones. (1 mark)

_____

(c) Explain why skeletal muscles occur in antagonistic pairs. (2 marks)

_____

_____

##

(a) (i) R (1)

(ii) P (1)

(b) Tendons (1)

(c) Skeletal muscles cannot lengthen by themselves after contracting. (1)

They need opposing muscles to pull the bone back to the starting position. (1)

(5 marks)

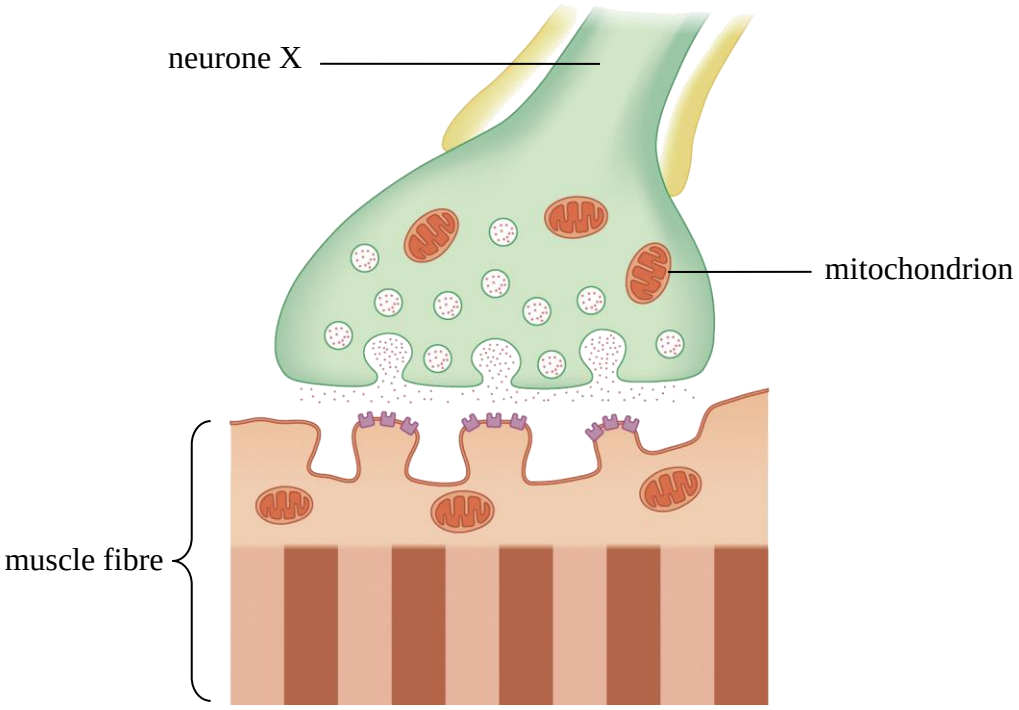
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##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

|!|BCESB0180000011|!| ★ **Cross topic**

The diagram below shows a neuromuscular junction.



(a) What type of neurone is X? Explain your answer. (2 marks)

_____

_____

(b) Identify and label ***two*** features in the above diagram that are involved in the transmission of nerve impulse and are also found in a synapse. (2 marks)

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(c) Describe how nerve impulses can be transmitted across the neuromuscular junction to trigger muscle contraction. (3 marks)

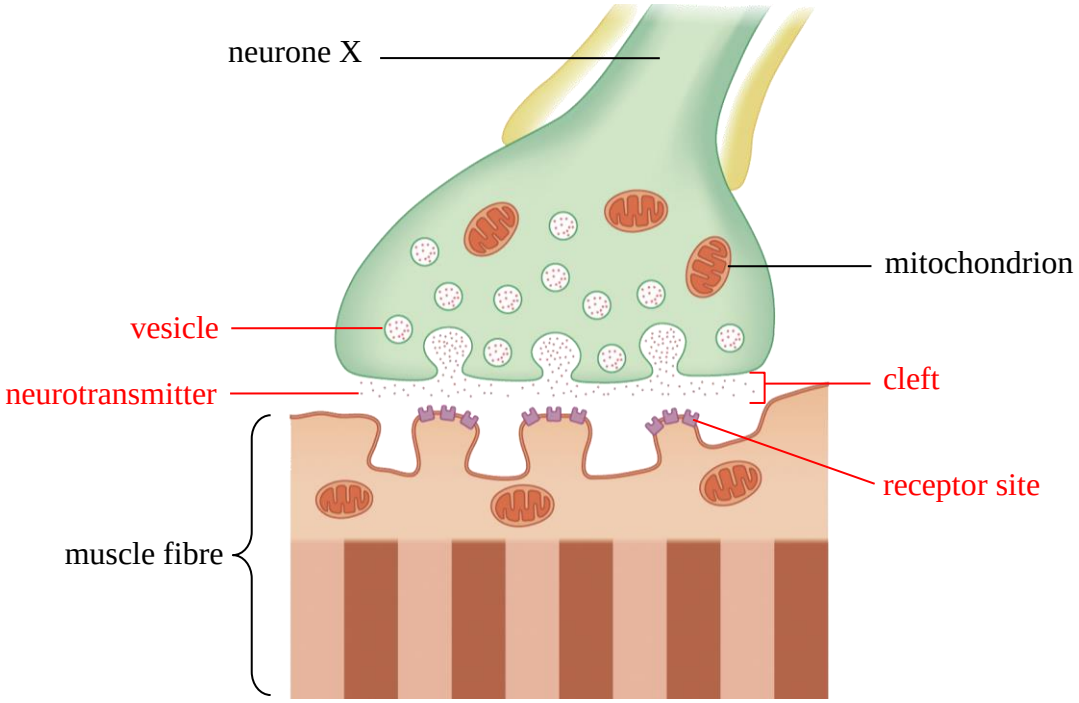
##

(a) Motor neurone (1)

It is connected to an effector (i.e. muscle fibre in this case). (1)

(b) Correct identification and labelling of any ***two*** of the following: (2)

- vesicle /
- neurotransmitter /
- cleft /
- receptor site



(c) The arrival of nerve impulses at the motor nerve ending triggers the release of neurotransmitters into the neuromuscular junction. (1)

These neurotransmitters diffuse across the synaptic cleft (1)

and bind to the receptor side on the membrane of the muscle fibre (1)

to trigger muscle contraction.

(7 marks)

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##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Presenting information	2
(c)	Understanding basic concepts	3

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